

- 15 Earth has a mass M and radius R . X is a point $5R$ from the center of the Earth.
An object of mass m falls freely from rest at X and hits the surface of the Earth.
Which of the following statements is false?

- A The change in gravitational potential is $\frac{4}{5} \frac{GM}{R}$.
B The work done by the gravitational field is $\frac{4}{5} \frac{GMm}{R}$.
C The speed of impact is $\sqrt{\frac{8}{5} \frac{GM}{R}}$.
D The change in the magnitude of gravitational field strength is $\frac{24}{25} \frac{GM}{R^2}$.

Ans: A

Option A is an incorrect statement because as the object is falling, it should lose gravitational potential and hence the change should take on a negative value.

$$\begin{aligned}\text{Change in gravitational potential} &= \phi_{\text{final}} - \phi_{\text{initial}} \\ &= -\frac{GM}{R} - \left(-\frac{GM}{5R} \right) \\ &= -\frac{4GM}{5R}\end{aligned}$$

Option B is correct statement because the direction of gravitational force and displacement of object is the same. Hence work done by gravitational field is positive. The amount of work done by gravitational field will be the same as the amount of GPE loss in the fall.

Option C is correct statement because
Loss in GPE = Gain in KE

$$\text{GPE}_{\text{initial}} - \text{GPE}_{\text{final}} = \text{KE}_{\text{final}} - \text{KE}_{\text{initial}}$$

$$-\frac{GMm}{5R} - \left(-\frac{GMm}{R} \right) = \frac{1}{2}mv^2$$

$$\frac{4GMm}{5R} = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{8GM}{5R}}$$

Option D is correct

$$\begin{aligned}\text{Change in gravitational field strength} &= g_{\text{final}} - g_{\text{initial}} \\ &= \frac{GM}{R^2} - \left(\frac{GM}{(5R)^2} \right) \\ &= \frac{24GM}{25R^2}\end{aligned}$$

16 The diagram shows the displacement-time graphs of two sound waves P and Q at a point in